



DEMOSAT

HIGH ALTITUDE BALLOON PAYLOAD



REQUIREMENTS

01

Under 200g

Total weight limit including all electronics

02

Under \$147

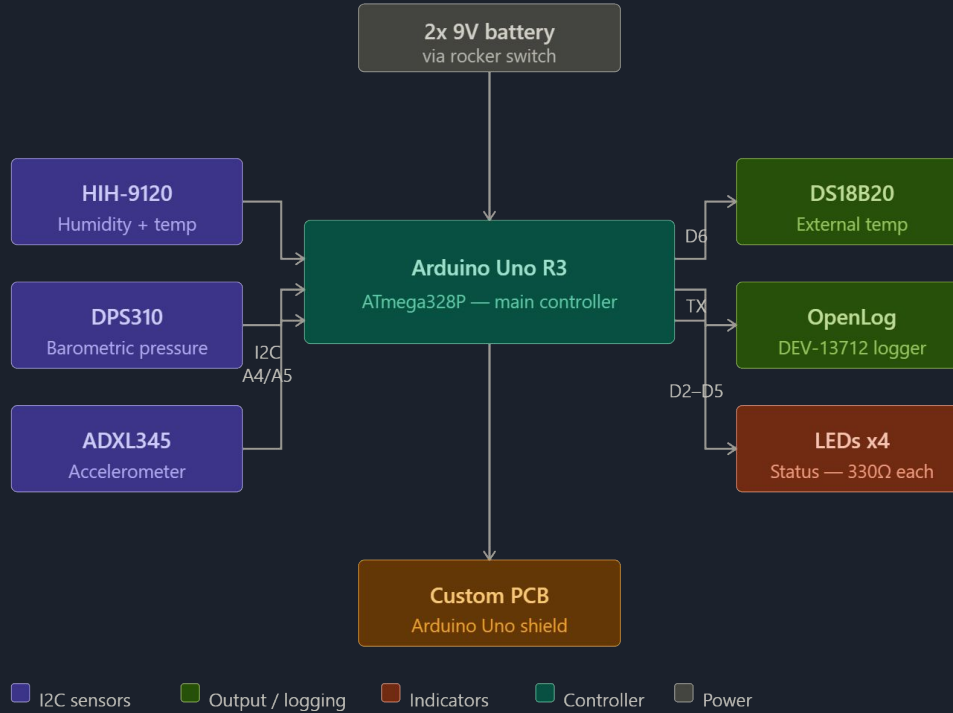
Total budget for all components

03

Less power than original

Reduce current draw compared to original design

Block Diagram



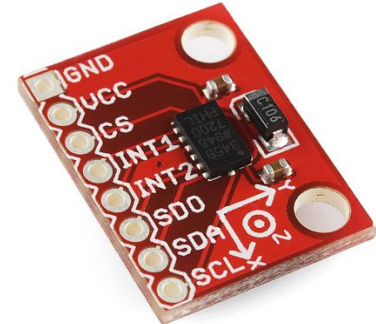
ACCELEROMETER — ADXL345

ADXL345

Analog Devices | Adafruit #1231 | \$9.95

- Supports I2C — works on SDA/SCL
- Higher resolution: 13-bit vs 10-bit
- Range: $\pm 2g$ / $\pm 4g$ / $\pm 8g$ / $\pm 16g$ selectable
- Lower power: $\sim 140\mu A$ @ 3.3V
- Excellent Arduino library support

ADXL345



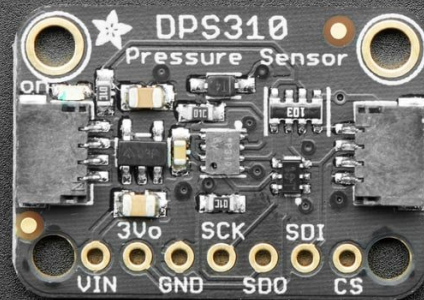
PRESSURE SENSOR — DPS310

DPS310

Infiniteon | Adafruit #4494 | \$6.95

- Supports I2C — address 0x77
- Pressure range: 300–1200 hPa
- Runs on 1.7–3.6V (vs 5V original)
- Ultra low power: $\sim 1.7\mu\text{A}$ @ 3.3V
- Compact form factor

DPS310



HUMIDITY SENSOR — HIH-9120-021-001

HIH-9120

Honeywell | Digi-Key | \$12.00

- Supports I2C — address 0x27
- Runs on 3.3V
- Accuracy: $\pm 1.7\%$ RH (vs $\pm 3.5\%$ original)
- Much smaller and lighter design
- $\sim 565\mu\text{A}$ @ 3.3V

HIH-9120



TEMPERATURE SENSOR — DS18B20

DS18B20

Dallas Semi | Adafruit #381 | \$4.95

- Digital output — no ADC required
- Resolution: 9–12 bit selectable
- Accuracy: $\pm 0.5^{\circ}\text{C}$ (vs $\pm 2^{\circ}\text{C}$ original)
- Waterproof probe for external sensing
- Still in production — unchanged

DS18B20



© Photo by ElectroPeak



TOTAL POWER, PRICE, AND WEIGHT

All requirements met

~103g

Total Weight

under 200g limit

~\$115

Total Cost

under \$147 budget

259mW

Typical Power

459mW with all LEDs on

0

Discontinued Parts

fully sourceable

~13hrs

Battery Life

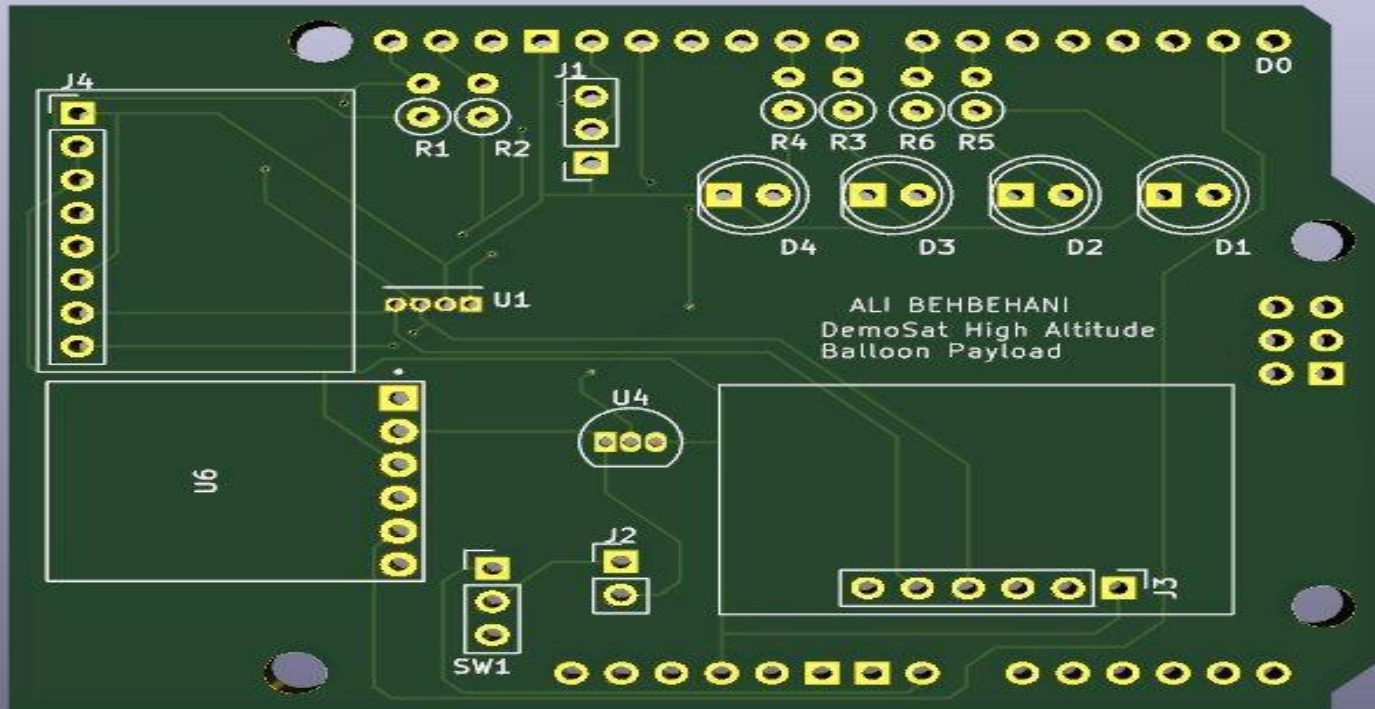
2x 9V alkaline

Lower

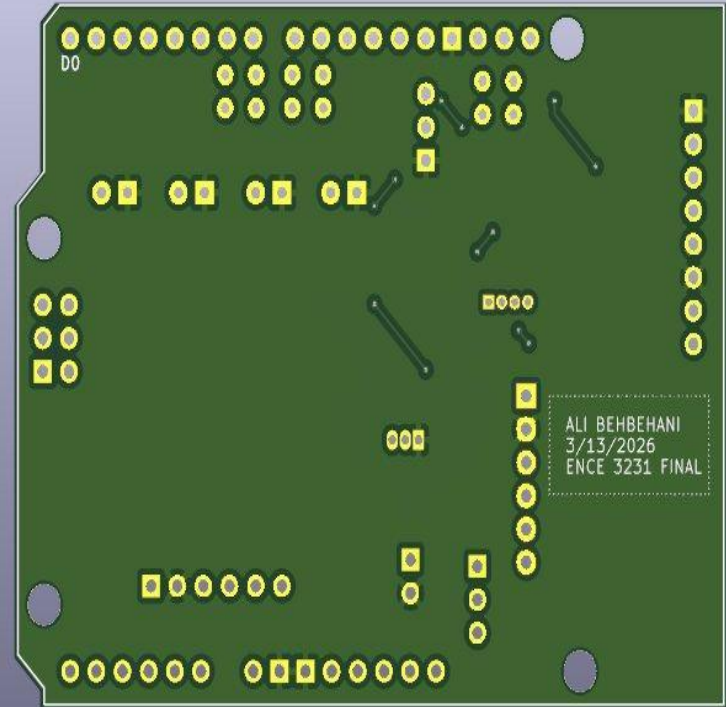
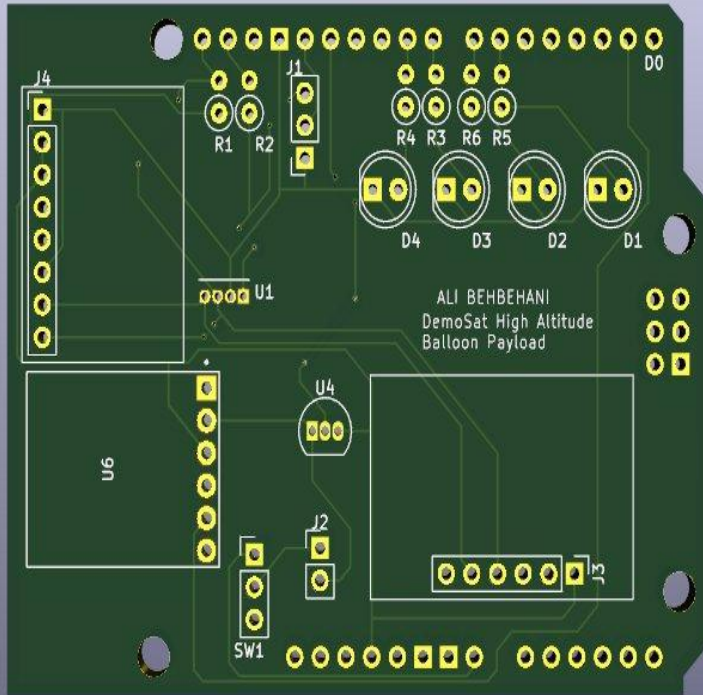
Power vs Original

digital sensors more efficient

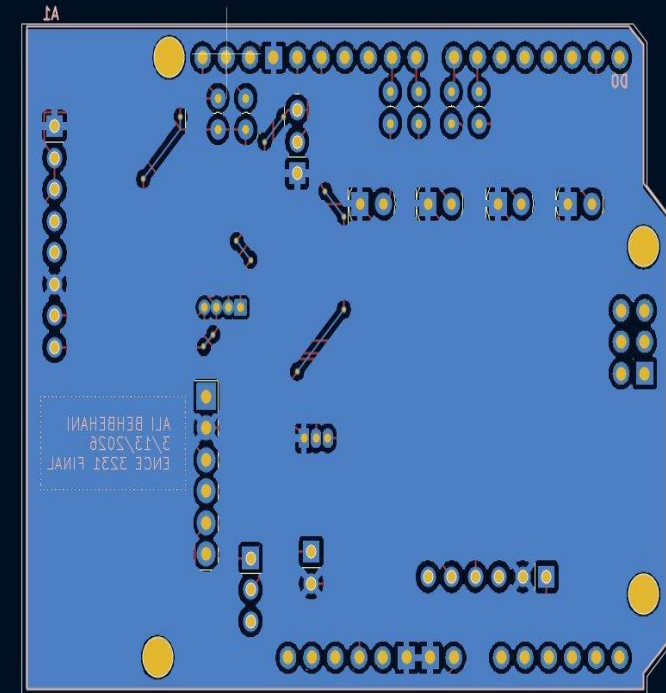
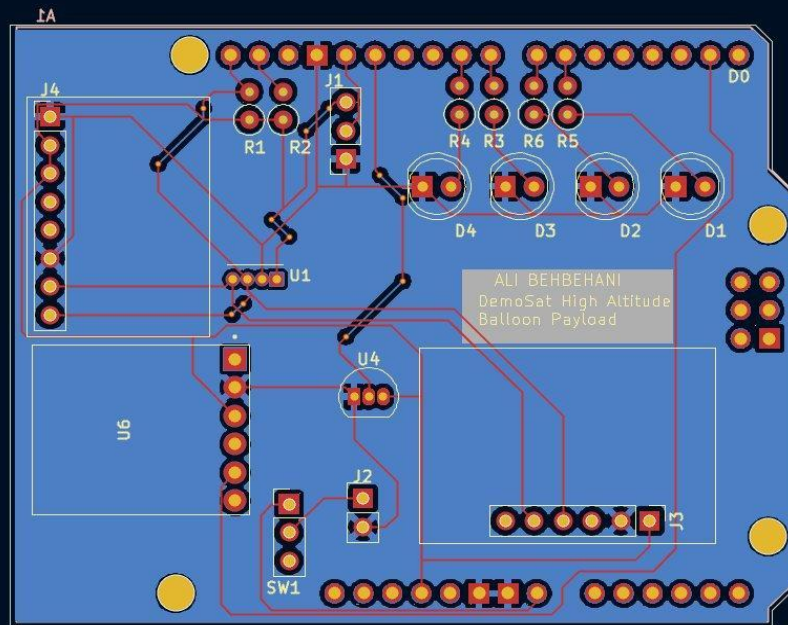
3D MODEL



FRONT / BACK — 3D RENDER



PCB LAYOUT — KICAD VIEW





BOM

#	Component	Part	Price	Weight
1	Microcontroller	Arduino Uno R3	\$28.00	25g
2	Humidity + Temp	HIH-9120-021-001	\$12.00	2g
3	Pressure	DPS310 (Adafruit #4494)	\$6.95	2g
4	Accelerometer	ADXL345 (Adafruit #1231)	\$9.95	2g
5	External Temp	DS18B20 Waterproof Probe	\$4.95	8g
6	Data Logger	OpenLog DEV-13712	\$24.95	10g
7	LEDs + Resistors	4x LED + 4x 330Ω + 3x 4.7kΩ	\$2.35	1g
8	Power	2x 9V Battery + Switch + Jack	\$9.50	95g
9	Custom PCB	2-Layer Shield 69.5×54mm	\$10.00	12g
TOTAL			~\$115	~157g



SUMMARY

0 discontinued parts — fully sourceable

~\$115 total — under \$147 budget

~157g — under 200g weight limit

~13 hour battery life — covers 3–4 hour flight

Custom KiCad PCB shield — 2-layer, 69.5×54mm

All sensors use I2C or 1-Wire — clean wiring